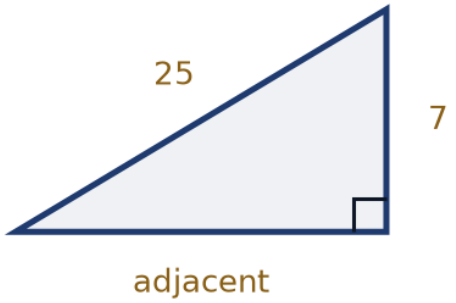
Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

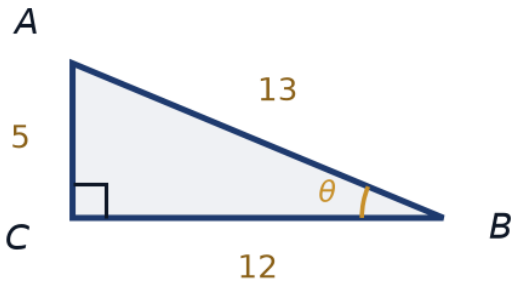
1 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, the angle θ has an opposite side of 7 and hypotenuse 25. Find $\cos\theta$ and $\tan\theta$.



2 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, angle θ has an adjacent side of 8 and opposite side 15. Find $\sin\theta$ and the hypotenuse.



3 [Foundational] • Target: Exact values and the unit circle

State the exact value of $\sin\left(\frac{\pi}{3}\right)$ and $\cos(45^\circ)$.

4 [Foundational] • Target: Exact values and the unit circle

State the exact value of $\tan(60^\circ)$ and $\sin(30^\circ)$.

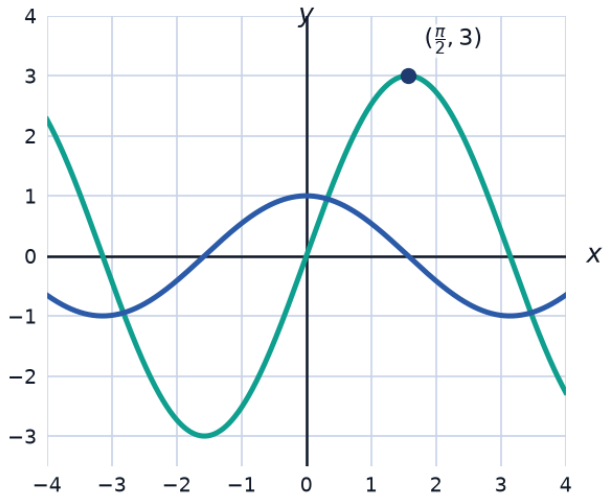
5 [Foundational] • Target: Graphs of trigonometric functions

State the amplitude and period of $y = 4\sin(2x)$.

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6 [Foundational] • Target: Graphs of trigonometric functions

State the amplitude and period of $y = 3\cos\left(\frac{x}{2}\right)$.



7 [Foundational] • Target: Trigonometric identities

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using the identity $\sin^2\theta + \cos^2\theta = 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Trigonometric identities

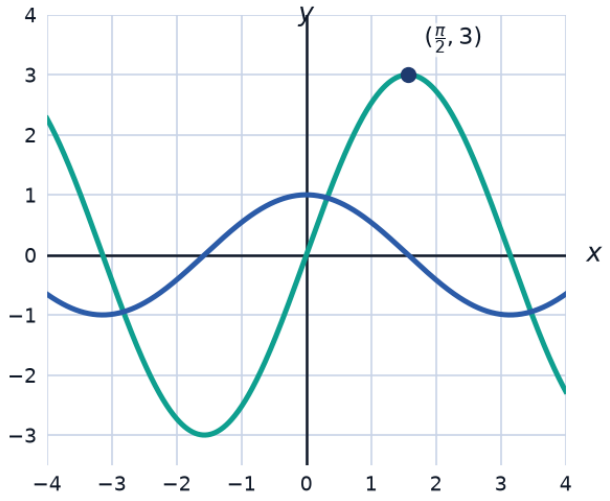
Simplify $\frac{\sin\theta}{1-\cos\theta} \times (1 - \cos\theta)$.

9 [Intermediate] • Target: Trigonometric identities

Given $\cos\theta = \frac{5}{13}$ with θ acute, find $\sin\theta$ using the identity $\sin^2\theta + \cos^2\theta = 1$.

10 [Intermediate] • Target: Solving trigonometric equations

Solve $2\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$.



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11 [Intermediate] • Target: Solving trigonometric equations

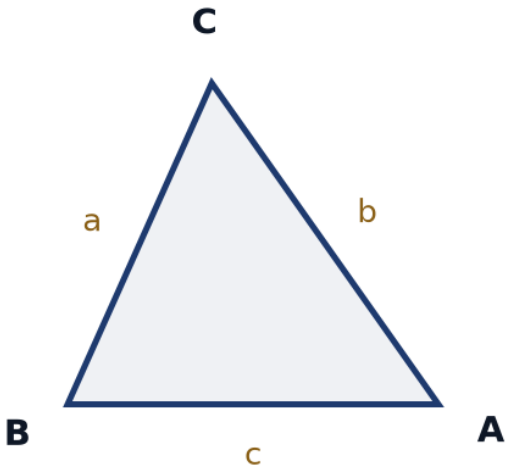
Solve $\cos(2\theta) = -1$ for $0^\circ \le \theta < 360^\circ$.

12 [Intermediate] • Target: Solving trigonometric equations

Solve $\tan\theta = -1$ for $0^\circ \le \theta < 360^\circ$.

13 [Intermediate] • Target: Law of Sines and Law of Cosines

In triangle ABC, $a = 9$, $b = 12$, and angle $C = 55^\circ$. Find side c, to the nearest tenth, using the Law of Cosines.



14 [Intermediate] • Target: Law of Sines and Law of Cosines

In triangle ABC, angle $A = 40^\circ$, angle $B = 70^\circ$, and $a = 8$. Find side b, to the nearest tenth, using the Law of Sines.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

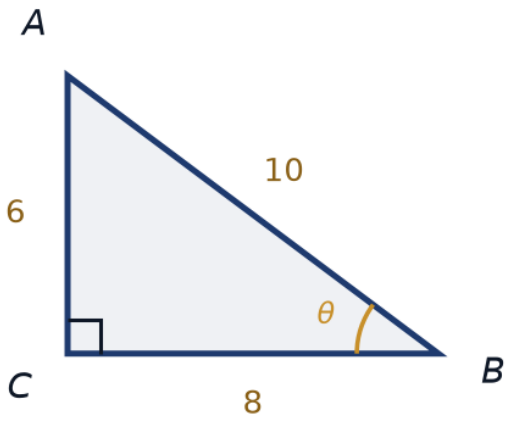
15 [Challenging] • Target: Law of Sines and Law of Cosines • Cross-Lesson: 19.01 Normal Distribution

In triangle ABC, $a = 7$, $b = 10$, $c = 13$. Find the largest angle, to the nearest degree, using the Law of Cosines.

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16 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Use the identity $\sin(A+B)=\sin A\cos B+\cos A\sin B$ to find the exact value of $\sin(75^\circ)$, using $75^\circ=45^\circ+30^\circ$.



17 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 3.01 Graphs & Transformations

Use the identity $\cos(A-B)=\cos A\cos B+\sin A\sin B$ to find the exact value of $\cos(15^\circ)$, using $15^\circ=45^\circ-30^\circ$.

18 [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 4.01 Coordinate Geometry

This question has two parts. A ship observes a lighthouse at an angle of elevation of 18°. After sailing 500 meters directly toward the lighthouse, the angle of elevation increases to 30°. (a) Let h be the lighthouse's height and d the final distance from its base. Write two equations relating h and d using the tangent ratio. (b) Solve the system to find h, to the nearest meter.

19 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 6.01 Differentiation & Integration

Describe fully the single transformation that maps $y=x^2$ to $y=\left(\frac{x}{3}\right)^2$.

20 [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 7.01 Statistics & Probability

Describe fully the single transformation that maps $y=x^2$ to $y=\left(\frac{x}{4}\right)^2$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exact Trig Ratios & Reference Angles

Find the exact value of $\sin(210^\circ)$.

a. $-\frac{1}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. $\frac{\sqrt{3}}{2}$

22 [Foundational] • Target: Exact Trig Ratios & Reference Angles

Find the exact value of $\cos\left(\frac{3\pi}{4}\right)$.

a. $-\frac{\sqrt{2}}{2}$

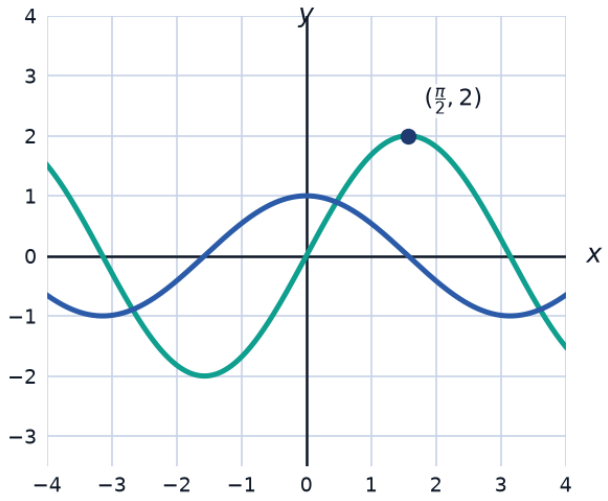
b. $\frac{\sqrt{2}}{2}$

c. $-\frac{1}{2}$

d. $\frac{\sqrt{3}}{2}$

23 [Foundational] • Target: Exact Trig Ratios & Reference Angles

If θ is reflex and $\tan(\theta) = \frac{4}{3}$, find the exact value of $\sin(\theta)$.



a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $\frac{3}{5}$

24 [Foundational] • Target: Fundamental Trigonometric Identities

Simplify the expression $(1 - \cos^2\theta)(1 + \cot^2\theta)$ for all valid θ.

a. 1

b. $\sin^2\theta$

c. $\tan^2\theta$

d. $\sec^2\theta$

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25 [Foundational] • Target: Fundamental Trigonometric Identities

Prove the identity $\frac{\sin\theta}{1-\cos\theta} + \frac{1-\cos\theta}{\sin\theta} \equiv 2\csc\theta$. What is the common numerator after combining fractions?

a. $2(1-\cos\theta)$

b. $\sin^2\theta + \cos^2\theta$

c. $2\sin\theta$

d. $1 - 2\cos\theta$

26 [Foundational] • Target: Fundamental Trigonometric Identities

Given $\cos\theta = -\frac{5}{13}$ and $90^\circ < \theta < 180^\circ$, find $\tan\theta$.

a. $-\frac{12}{5}$

b. $\frac{12}{5}$

c. $-\frac{5}{12}$

d. $\frac{13}{12}$

27 [Foundational] • Target: Solving Trigonometric Equations

Solve $2\sin(2x) = 1$ for $0^\circ \le x \le 180^\circ$.

a. $x = 15^\circ$ and $x = 75^\circ$

b. $x = 30^\circ$ and $x = 150^\circ$

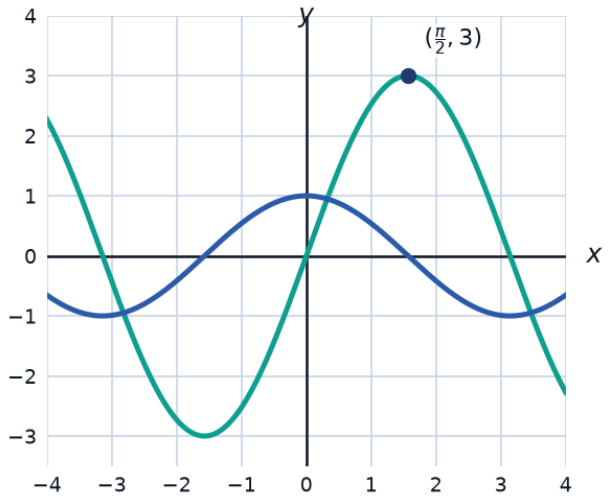
c. $x = 15^\circ$ and $x = 105^\circ$

d. $x = 45^\circ$ and $x = 135^\circ$

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28 [Foundational] • Target: Solving Trigonometric Equations

Solve $2\cos^2\theta - 5\cos\theta + 2 = 0$ for $0^\circ \le \theta \le 360^\circ$.



a. $\theta = 60^\circ$ and $\theta = 300^\circ$

b. $\theta = 30^\circ$ and $\theta = 330^\circ$

c. $\theta = 60^\circ$ and $\theta = 120^\circ$

d. $\theta = 45^\circ$ and $\theta = 315^\circ$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Solving Trigonometric Equations

Solve $3\tan(x) = 2\sin(x)$ for $0 < x < 2\pi$.

a. $x = \pi$ (excluded) or $\cos x = 3/2$ (no solution)

b. No solutions in $(0, 2\pi)$

c. $x = \pi/3$ and $5\pi/3$

d. $x = \pi/6$ and $5\pi/6$

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30 [Intermediate] • Target: Trigonometric Graphs & Transformations

State the amplitude and period of the function $y = 3\sin(2x) - 1$.

a. Amplitude $= 3$, Period $= \pi$

b. Amplitude $= 3$, Period $= 2\pi$

c. Amplitude $= 2$, Period $= \pi$

d. Amplitude $= 4$, Period $= \pi$

31 [Intermediate] • Target: Trigonometric Graphs & Transformations

Find the range of the function $f(x) = 4 - 3\cos(x)$ for all real x.

a. $[1, 7]$

b. $[-1, 7]$

c. $[1, 4]$

d. $[-3, 3]$

32 [Intermediate] • Target: Trigonometric Graphs & Transformations

Find the number of solutions to the equation $\sin(2x) = \cos(x)$ in the interval $0 \le x \le 2\pi$.

a. 4 solutions

b. 2 solutions

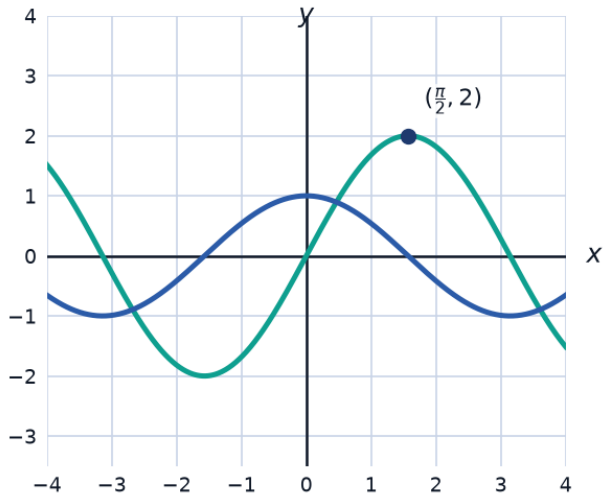
c. 3 solutions

d. 6 solutions

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33 **[Intermediate]** • **Target: Right-triangle trigonometry**

Describe fully the single transformation that maps $y = x^2$ to $y = \left(\frac{x}{2}\right)^2$.



a. Stretch parallel to the x-axis, scale factor $\frac{1}{2}$

b. Stretch parallel to the x-axis, scale factor 2

c. Stretch parallel to the y-axis, scale factor 2

d. Translation parallel to the x-axis

34 **[Intermediate]** • **Target: Exact values and the unit circle**

The minimum point of $y = x^2$ is translated by the vector $(-3; 6)$ to give curve C. State the coordinates of the minimum point of C.

a. (3,6)

b. (6,-3)

c. (-3,6)

d. (-3,-6)

35 **[Intermediate]** • **Target: Graphs of trigonometric functions**

The minimum point of $y = x^2$ is translated by the vector $(-6; 5)$ to give curve C. State the coordinates of the minimum point of C.

a. (-6,-5)

b. (5,-6)

c. (-6,5)

d. (6,5)

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36 [Intermediate] • Target: Trigonometric identities

The minimum point of $y = x^2$ is translated by the vector $(1;\ -1)$ to give curve C. State the coordinates of the minimum point of C.

a. (-1,1)

b. (1,1)

c. (1,-1)

d. (-1,-1)

37 [Intermediate] • Target: Solving trigonometric equations

The minimum point of $y = x^2$ is translated by the vector $(6;\ -5)$ to give curve C. State the coordinates of the minimum point of C.

a. (6,-5)

b. (-6,-5)

c. (6,5)

d. (-5,6)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Law of Sines and Law of Cosines • Cross-Lesson: 4.01 Coordinate Geometry

The minimum point of $y = x^2$ is translated by the vector $(-2; 3)$ to give curve C. State the coordinates of the minimum point of C.

a. (-2,3)

b. (-2,-3)

c. (3,-2)

d. (2,3)

39 [Challenging] • Target: Trigonometric identities for compound angles • Cross-Lesson: 6.01 Differentiation & Integration

The minimum point of $y = x^2$ is translated by the vector $(1;\ -7)$ to give curve C. State the coordinates of the minimum point of C.

a. (-1,-7)

b. (1,-7)

c. (-7,1)

d. (1,7)

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40 [Challenging] • Target: A well-designed word problem: the height of a lighthouse • Cross-Lesson: 7.01 Statistics & Probability

The minimum point of $y = x^2$ is translated by the vector $(-3; 1)$ to give curve C. State the coordinates of the minimum point of C.

a. (1,-3)

b. (-3,-1)

c. (-3,1)

d. (3,1)

41 [Challenging] • Target: Exact Trig Ratios & Reference Angles • Cross-Lesson: 8.01 Functions: Composite & Inverse

The minimum point of $y = x^2$ is translated by the vector $(4; -4)$ to give curve C. State the coordinates of the minimum point of C.

a. (-4,4)

b. (4,4)

c. (4,-4)

d. (-4,-4)

42 [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 9.01 Circular Measure

The minimum point of $y = x^2$ is translated by the vector $(7; -6)$ to give curve C. State the coordinates of the minimum point of C.

a. (7,6)

b. (-6,7)

c. (-7,-6)

d. (7,-6)

43 [Challenging] • Target: Solving Trigonometric Equations • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 2, then translated by $(4; -7)$. State the coordinates of the new vertex.

a. (-4,-7)

b. (4,-7)

c. (-7,4)

d. (4,7)

44 [Challenging] • Target: Trigonometric Graphs & Transformations • Cross-Lesson: 10.01 Series & Binomial Theorem

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(1; -6)$. State the coordinates of the new vertex.

a. (-1,-6)

b. (1,-6)

c. (1,6)

d. (-6,1)

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45 [Challenging] • Target: Right-triangle trigonometry • Cross-Lesson: 11.01 Kinematics

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 2, then translated by $(7;\ -4)$. State the coordinates of the new vertex.

a. (7,-4)

b. (7,4)

c. (-7,-4)

d. (-4,7)

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